StillCold — Week 7 Progress

Flutter Companion App MVP

Building the mobile side of the StillCold system

What Week 7 Was About

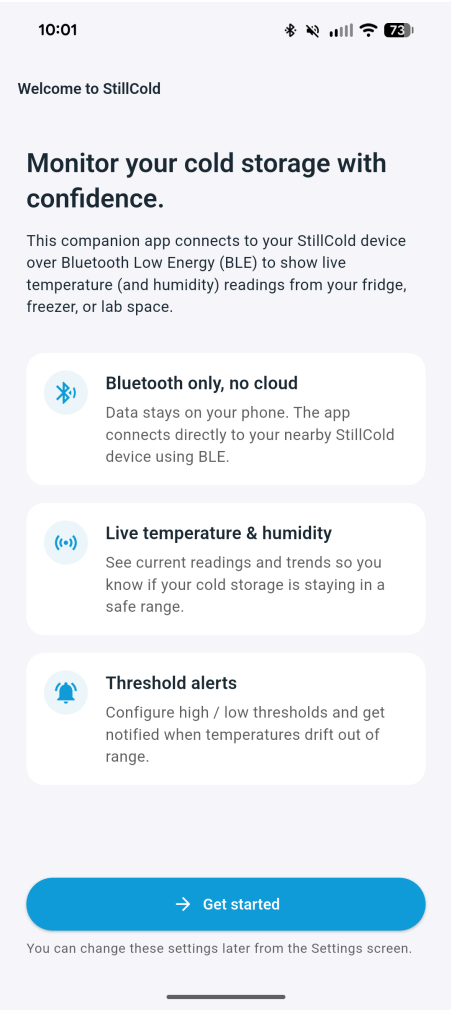
Goal: Move from BLE-only hardware validation to a working **Flutter companion app** that:

- Discovers **StillCold** devices over BLE.
- Connects and reads **live temperature and humidity**.
- Shows readings in a **modern, user-friendly UI**.
- Implements **threshold alerts, quiet hours, and alert history**.
- Begins aligning the implementation to the **formal SRS**.

First-Run Experience

The app now has an **Onboarding screen** that introduces the companion app and sets expectations:

- Explains that the app uses **Bluetooth Low Energy (BLE)**, not the cloud.
- Emphasizes **live temperature & humidity monitoring**.
- Highlights **threshold alerts** as a core feature.
- Guides the user to **Get started** and proceed to discovery.


10:01
73
Welcome to StillCold
Monitor your cold storage with confidence.
This companion app connects to your StillCold device over Bluetooth Low Energy (BLE) to show live temperature (and humidity) readings from your fridge, freezer, or lab space.
Bluetooth only, no cloud
Data stays on your phone. The app connects directly to your nearby StillCold device using BLE.
Live temperature & humidity
See current readings and trends so you know if your cold storage is staying in a safe range.
Threshold alerts
Configure high / low thresholds and get notified when temperatures drift out of range.
Get started
You can change these settings later from the Settings screen.

Device Discovery Over BLE

The app now acts as a **Bluetooth “radar”** for nearby StillCold devices:

- Scans for nearby devices and only shows those named **"StillCold"**.
- Clearly explains what’s happening while scanning.
- Gives simple, human-readable messages if:
  - No devices are found.
  - The phone is missing a required permission.
  - Something goes wrong during the scan.
- Shows **signal strength** for each device, so you know which one is closest.
- Remembers which device you used last, and surfaces that in the discovery view.

From here, a single tap takes you into the **Dashboard** for that device.

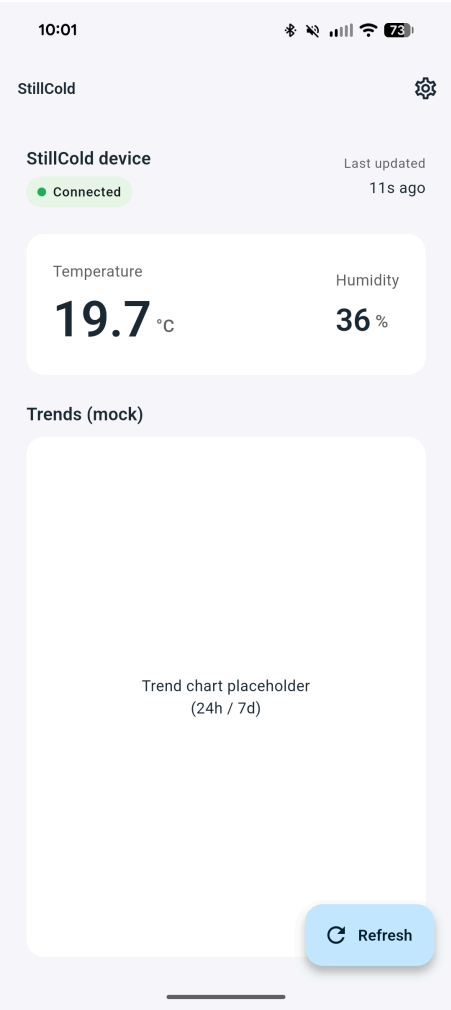
Dashboard: Live Readings

The **Dashboard** presents the core environmental data:

- **Connection status** chip (connected).
- Prominent **temperature** card with one decimal precision.
- **Humidity** percentage.
- **Last updated** indicator (e.g., “Just now”, “11s ago”).
- **Refresh** button to manually request a new reading.

Each reading:

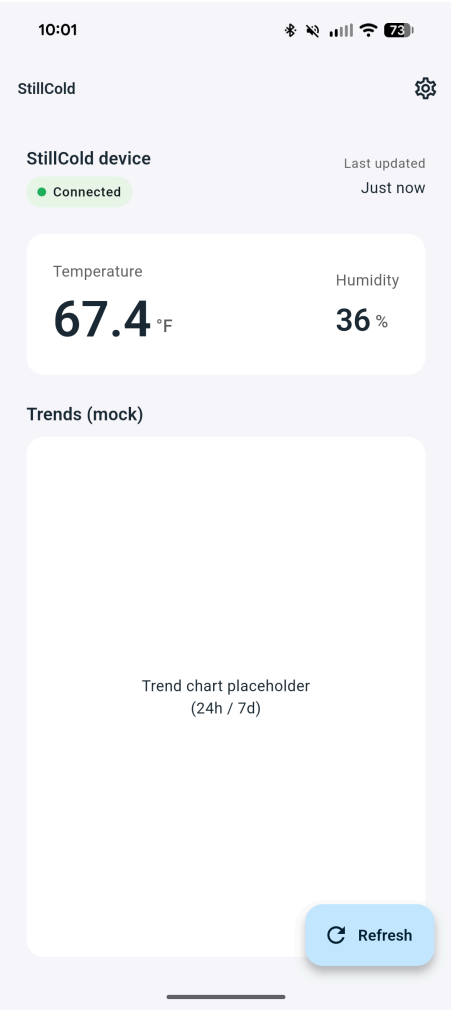
- Comes directly from the StillCold device over Bluetooth.
- Is tagged with **when** it was taken.
- Is saved on the phone so that history and alerts can be reviewed later.


10:01
73
StillCold
StillCold device
Connected
Last updated
11s ago
Temperature
19.7 °C
Humidity
36 %
Trends (mock)
Trend chart placeholder
(24h / 7d)
Refresh

Temperature Units: °C / °F

The app supports user preference for **Celsius or Fahrenheit**:

- **Settings** screen exposes a **“Use Fahrenheit”** toggle.
- The Dashboard:
  - Converts temperature as needed.
  - Updates both value **and unit label** consistently.
- Humidity display remains unchanged.


10:01
73
StillCold
StillCold device
Connected
Last updated
Just now
Temperature
67.4 °F
Humidity
36 %
Trends (mock)
Trend chart placeholder
(24h / 7d)
Refresh

Thresholds, Quiet Hours, and Alerts

On top of raw readings, the app now helps answer the question:
"Is my cold storage staying within a safe range?"

- **Custom thresholds**
  - Set your own safe **low** and **high** temperature limits.
  - Defaults are chosen to match typical refrigeration guidance.
- **Quiet hours**
  - Optional "do not disturb" window (for nights or meetings).
  - Alerts are silently logged during this time without buzzing your phone.

- **Alert history**
  - A simple timeline showing **when things went out of range**, and by how much.
  - Lets you see patterns (e.g., "door left open every evening around 6pm").
- **On-device notifications**
  - When the temperature crosses your thresholds (and you're not in quiet hours), the phone shows a clear notification so you can act quickly.

Data and Privacy: Where Information Lives

All companion-app data **stays on the phone**:

- Temperature and humidity readings are stored locally for history and future charts.
- Your preferences (units, thresholds, quiet hours, last device) are also stored locally.
- Alert history never leaves the device.

There is:

- **No cloud backend.**
- **No internet requirement.**

If your phone can talk to the StillCold device over Bluetooth, you get full functionality today, and the design leaves room to grow with charts, multi-device support, and more.

Alignment with the SRS

A dedicated document, `SRS_status_week7.md`, now tracks **requirement status**:

- **Most Sprint 1 Must-have FRs** are implemented:
  - BLE discovery and connection.
  - Live temperature/humidity display.
  - Manual refresh + timestamped readings.
  - Local, on-device storage of readings and alerts.
  - Threshold configuration + alerts + quiet hours + alert history.
  - Unit toggling and basic UX/error states.
- Non-functional goals around **usability, maintainability, and Android compatibility** are largely met.
  This document provides a clear **traceability map** from SRS → codebase → current behavior.

What’s Still Missing (High Level)

From the SRS and status review:

- **Connection lifecycle polish**
  - Real-time connection state (connecting / connected / disconnected).
  - Explicit **Disconnect** action that returns to Discovery.
  - A dedicated **“Connection lost”** flow when BLE drops unexpectedly.

- **Sprint 2 enhancements**
  - Periodic **polling loop** tied to the configured interval.
  - Real **trend charts** (24h / 7d) using stored readings.
  - **Min/max summaries** (e.g., coldest/warmest last 24h).
  - **Dashboard RSSI** when connected.
  - **Device label editing UI** (add/rename/remove).
  - Visual indication when readings become **stale**.

These are the main targets for the next iteration.

Week 7 Summary

- The **StillCold Flutter companion app runs on real hardware** (Pixel 8) and:
  - Discovers the StillCold device over BLE.
  - Reads and displays live temperature and humidity.
  - Stores readings on the phone with timestamps.
  - Evaluates thresholds and surfaces alerts (including local notifications and history).
  - Lets the user configure units, thresholds, quiet hours, and basic device preferences.
- The implementation is now **firmly anchored to the SRS**, with an explicit status document for verification.

The project has effectively moved from **"BLE-capable hardware"** to a **working end-to-end system** with a mobile UI.

Thank you

StillCold — *Environmental monitoring without opening the door*

Questions?